

NOTEBOOK

OVERVIEW

ZJU-CHINA2026





Notebook — TVN Magnetic-Nanoparticle Binding Module

TVN / TVN–DogTag construct development, purification, fluorescent labelling and Fe_3O_4 interaction assays

Laboratory record: 22 June–28 September 2026. Reorganised and expanded from the original chronological TVN notebook. The observations below are preliminary unless validation is explicitly stated.

Timeline of the Work

Date (2026)	Work performed	Record
22 June–8 July	Initial TVN and control-peptide magnetic-particle binding tests; BCA and FITC pilot assays	1
9–12 July	Site-directed mutagenesis, transformation, screening and Sanger verification	2
13–16 July	IPTG induction screen, scale-up expression, His-tag purification and first FITC concentration screen	3
22–28 July	Dialysis, BCA quantification, ultrafiltration and SnoopLigase2-mediated DogTag ligation checks	4
2–4 August	FITC labelling procedure and magnetic-particle fluorescence optimisation	5
5–15 August	Repeat dialysis, concentration determination and fluorescence assay	6
28 August–4 September	Recovery of bacterial stocks and further protein production and quantification	7
16–28 September	Revised fluorescent-labelling conditions, SDS-PAGE controls and microscopy checks	8

1 Initial TVN adsorption assays and fluorescence feasibility tests

Date: 22 June–8 July 2026

Procedures

1. A synthetic TVN peptide (TVNFKLY) was compared with an unrelated control peptide (GTWSLLL). The Fe_3O_4 magnetic nanoparticles were recorded as approximately 20 nm in diameter.

2. For the initial depletion/BCA approach, nanoparticles were pretreated with BSA for 2 h and exposed to 500 µL of a 1 mg/mL peptide/protein solution at room temperature for 2 h. Particles were recovered on a magnetic rack and the remaining solution was tested by BCA.

3. On 24 June, the assay was repeated without the 2 h BSA pretreatment to determine whether surface blocking affected the signal.

4. On 8 July, an FITC-based binding pilot was performed using the protocol referenced in the original notebook ("FITC-labelled peptide and Fe_3O_4 nanoparticle binding without dialysis membrane").

Note. Both BCA trials produced an insufficiently discriminating readout; the early fluorescent pilot also showed no clearly visible labelling. Neither assay established quantitative binding efficiency. The notebook references a separate protocol file that was not supplied with these two documents.

2 Mutagenesis, transformation and sequence confirmation

Date: 9–12 July 2026

Procedures

1. Primers were designed and site-directed mutagenesis carried out following the Mut Express II Fast Mutagenesis Kit V2 workflow, including PCR amplification, Dpn I treatment and homologous recombination.

2. The reconstructed plasmid was introduced into Trelief 5α competent cells. Transformants were selected on kanamycin-containing agar at 37 °C overnight; plasmid was prepared using the SuperPlasmid Mini Kit and quantified with a NanoDrop instrument.

3. Candidate constructs were transferred to E. coli BL21(DE3) and examined by colony PCR and agarose electrophoresis using vector-backbone and insertion-site primer sets.

4. The first construction attempt failed on 10 July and was repeated. The reconstruction and subsequent transformation produced candidate colonies on 11–12 July.

5. Sanger sequencing was commissioned using the pET28a/PET-T7 universal sequencing primer. Two archived trace filenames were TVN-1.PET-T7.6093494.26-7-10J-4A06.F11.ab1 and TVN-2.PET-T7.6093495.26-7-10J-4A07.F12.ab1.

Note. The original notes report a 5,683-bp target construct and state that the Fe_3O_4-binding site was successfully introduced based on the sequencing comparison. The trace files and plasmid map are not present here, so the precise nucleotide changes cannot be independently rechecked in this notebook.

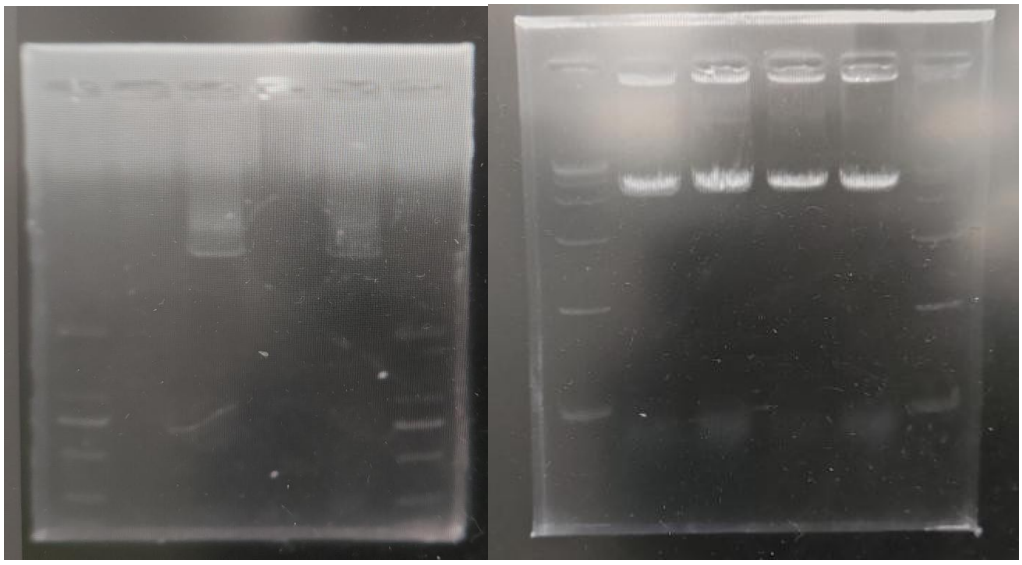


Figure. Agarose electrophoresis of the reconstructed plasmid; the original record labels the expected construct as 5,683 bp.

Figure. Colony-PCR gels for backbone- and insert-associated primer assays.

3 Recombinant expression, affinity purification and FITC screening

Date: 13–16 July 2026

Procedures

1. A single transformant was cultured in 10 mL of kanamycin-containing medium at 37 °C and 220 rpm for approximately 12 h, then diluted 1:100 and expanded to OD600 ≈ 0.6.

2. The IPTG induction screen used 0.25, 0.30, 0.42, 0.65 and 1.0 mM IPTG, followed by incubation at 30 °C and 200 rpm for 4 h. The 0.42 mM condition was noted as literature-associated rather than proven optimal in this experiment.

3. Whole-culture aliquots (20 µL) were combined with 5 µL sample buffer and heated at 95 °C for 15 min. Western blot samples were analysed with anti-His primary antibody (1:3,000), anti-GAPDH (1:10,000) and the corresponding HRP-conjugated secondary antibodies.

4. For native His-tag purification, bacterial pellets were resuspended in approximately 4 mL lysis buffer per gram wet mass. The recorded method used lysozyme (1 mg/mL; 30 min on ice), sonication (200–300 W; six 10 s pulses separated by 10 s), and clarification at 10,000 × g for 20–30 min at 4 °C.

5. Clarified lysate was bound to BeyoGold His-tag affinity resin, washed five times with 0.5–1 mL wash solution and eluted in six to ten separate 0.5 mL fractions. Lysate, flow-through, washes and eluates were retained for electrophoretic analysis.

6. A larger expression batch was performed on 14 July, followed by purification on 15 July and an SDS-PAGE fraction check on 16 July. An initial FITC gradient also compared 90, 215, 375 and 500 µg/mL input conditions, with the recorded particle input of 0.075 mg per group.

Note. The gradient images suggested that one intermediate-to-high fluorescence condition warranted further testing, but particle aggregation under the magnet obscured orientation or coverage. The operator proposed decreasing the particle input to 0.025 mg. Neither an expression-optimum calculation nor purified yield was established from the Western blot image alone.

Figure. Early FITC pilot micrograph; the contemporaneous note indicates that staining was not apparent.

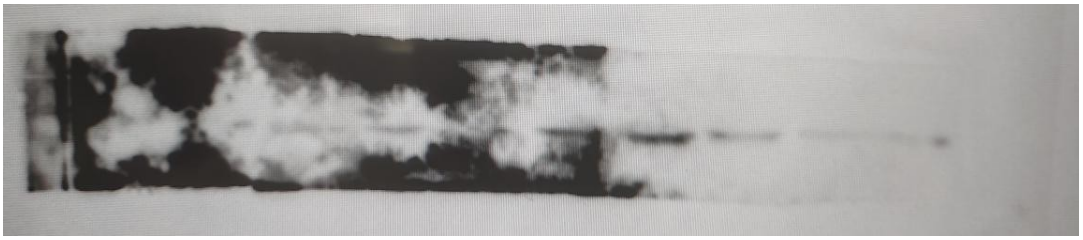


Figure. Anti-His Western blot image from the initial IPTG induction screen.

Figure. Representative FITC-labelled particle microscopy from the July screen; agglomeration reduced interpretability.

Figure. SDS-PAGE of lysate, flow-through, wash and elution samples collected during His-tag purification.

4 Dialysis, concentration recovery and DogTag ligation testing

Date: 22–28 July 2026

Procedures

1. Dialysis began on 22 July. The sample volume was recorded as increasing from approximately 4 mL to 10.5 mL by 23 July; aliquots were separated for storage.

2. BCA testing on 24 July indicated approximately 6.2 µg/mL for the diluted post-dialysis sample, corresponding to about 65.1 µg in 10.5 mL. A back-calculation in the original record estimated approximately 26 µg/mL for the earlier 2.5 mL sample volume.

3. For ligation assays, TVN–DogTag-containing fusion protein was incubated with GFP–SnoopTag2 and SnoopLigase2. Non-ligase reaction mixtures and SUMO–DogTag comparison mixtures were included in the planned lane layouts. Incubations of approximately 10–15 min, and later 6 h, were checked by SDS-PAGE.

4. Because the recombinant fusion was dilute, centrifugal ultrafiltration was carried out on 26 July. The construct was recorded as a 6×His–SUMO–DogTag–TVN arrangement. The sequence-derived mass was approximately 16.61 kDa, whereas a band migrated around 25 kDa in the gel. The 25 kDa apparent mass was used for the contemporaneous reaction-ratio estimate.

5. BCA measurements on 27–28 July assessed pre-concentration sample, retentate and filtrate. The reported calibration R^2 was 0.9940; the notebook lists approximately 0.374 mg/mL before concentration and 5.037 mg/mL afterward, with filtrate below the assay's detection threshold.

6. Ligation gels were repeated on 27 and 28 July after correcting sample loading and concentration differences. A planned particle-binding experiment was postponed because of a temporary water outage.

Reported fraction	BCA concentration	Interpretive caution
Before ultrafiltration	0.374 mg/mL	Recorded for the late-July batch
Concentrated retentate	5.037 mg/mL	Author noted dilution/retest may improve accuracy
Filtrate	Below limit of detection	Insufficient to quantify total mass recovery

Note. The 25 July experiment had inadvertent enzyme addition to two intended comparator lanes, an uneven resolving-gel interface and reused running buffer. Higher-molecular-weight bands were tentatively observed, but these limitations and unequal loading meant that the initial gel did not independently demonstrate specific ligation. Repeats improved the comparison but the source does not provide a quantified ligation yield. The strong discrepancy between predicted and apparent SDS-PAGE mass was documented, not resolved.

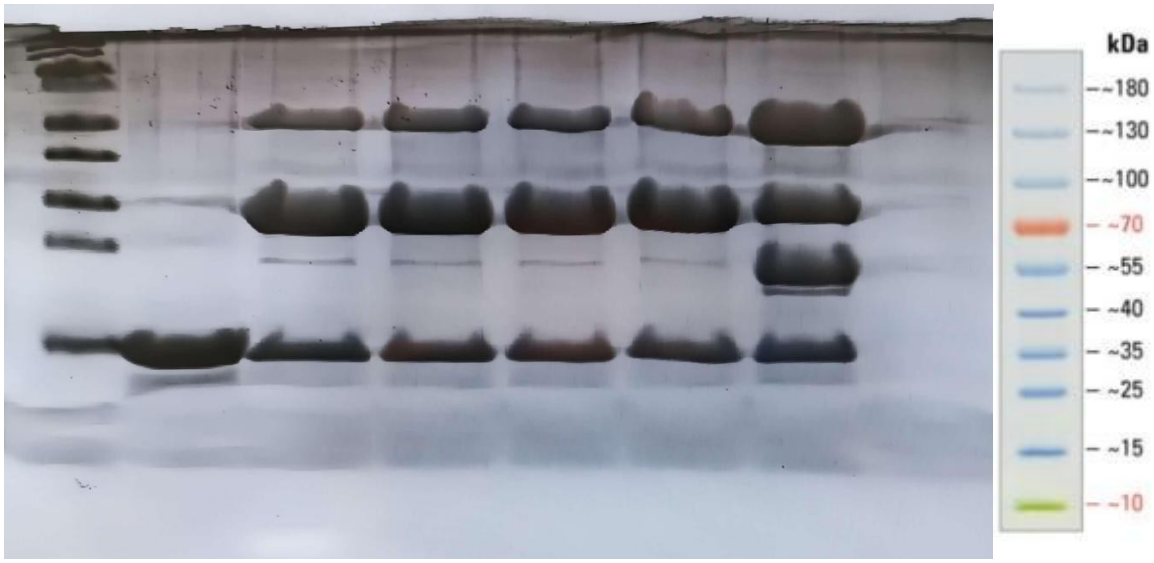


Figure. Initial DogTag/SnoopLigase2 ligation gel; its control assignment was compromised by sample-loading error.

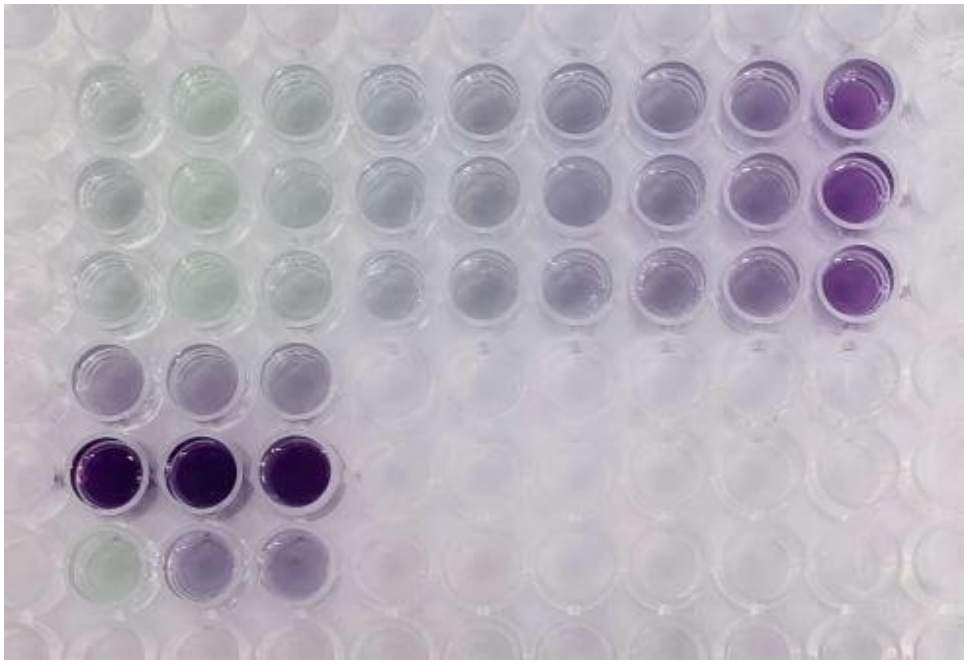


Figure. Follow-up BCA / reaction preparation record after ultrafiltration.

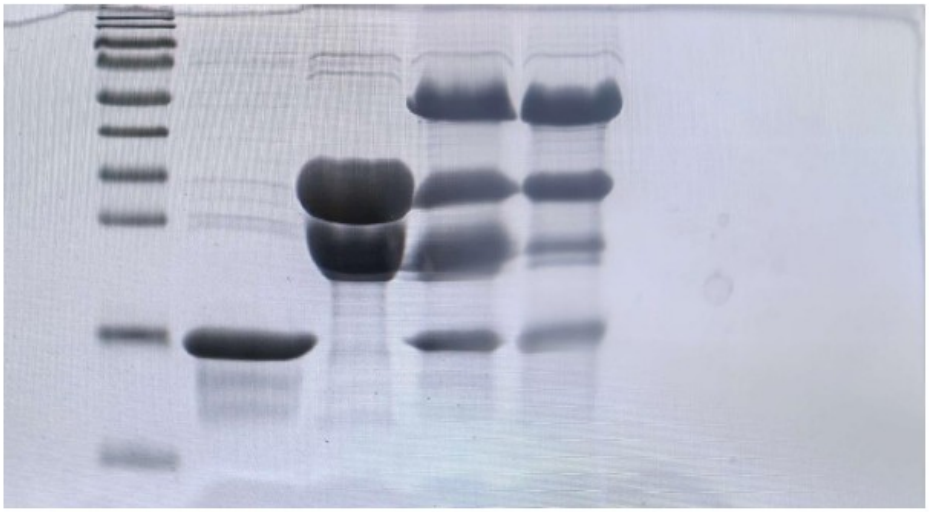


Figure. Follow-up SDS-PAGE image for the TVN–DogTag ligation assay.

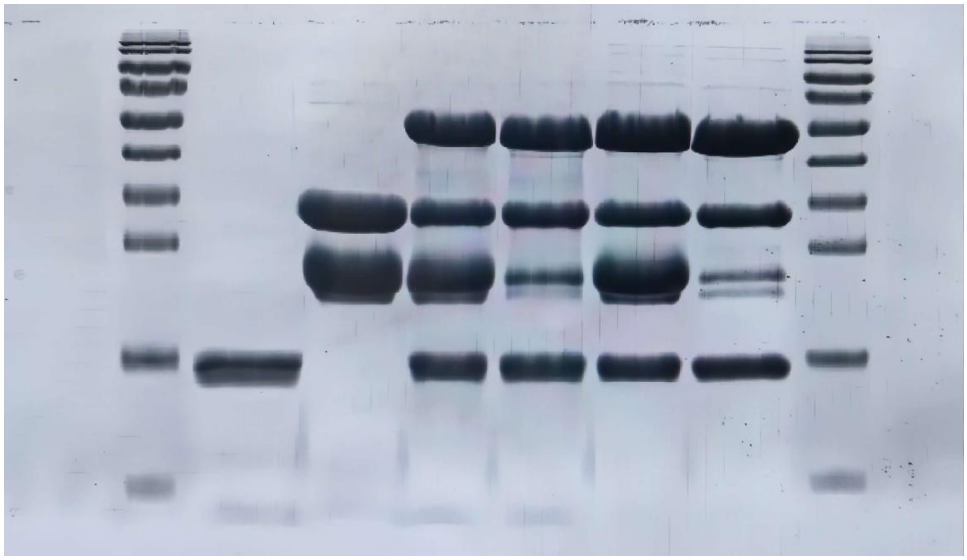


Figure. Repeat ligation activity gel comparing reporter, enzyme and DogTag-containing protein reaction mixtures.

5 FITC-labelling workflow and magnetic-particle interaction optimisation

Date: 2–4 August 2026

Procedures

1. The 2 August stock solutions were recorded as FITC in DMSO (10 mg/mL), 0.1 M carbonate buffer adjusted to pH 9.0, 1 M glycine and aqueous Fe_3O_4 nanoparticles. The carbonate formulation specified 0.84 g Na_2CO_3 and 0.19 g $NaHCO_3$ per 100 mL before final pH adjustment.

2. A 500 µL labelling system was specified using 50 µL of approximately 5.037 mg/mL fusion-protein stock, 10 µL of 10 mg/mL FITC, 415 µL carbonate buffer and 25 µL of 1 M glycine (added after the 2 h labelling step). FITC was added gradually, the reaction was protected from light, and free FITC was quenched for 30 min after the labelling period.

3. The fluorescence-labelled material was mixed with Fe_3O_4 particles and incubated for 2 h at room temperature in the dark, with gentle mixing every 30 min. A magnetic rack was used to isolate the particles and remove supernatant.

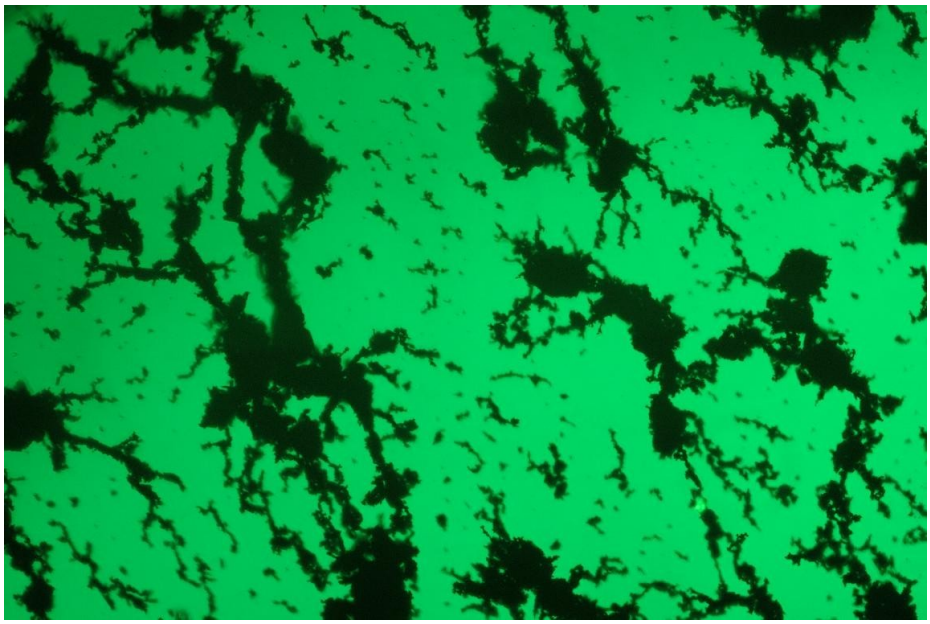
4. On 3 August microscopy was performed using the cellSens software and a fluorescein/FITC imaging setting. Only one bright-field magnet-condition image was retained because of a software issue. The operator noted weak binding and potential light exposure or excessive washing.

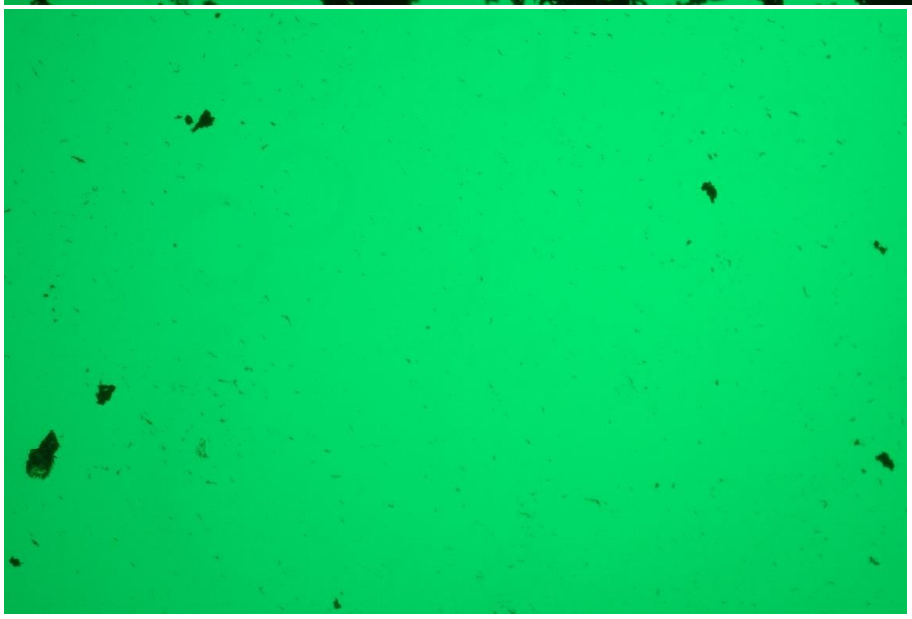
5. On 4 August controls and procedural variants compared FITC-only/particle-only preparations, synthetic TVN peptide, TVN–SUMO–DogTag fusion and unrelated GTW control, with and without glycine quench or sonication and with or without the magnet.

Component	Recorded volume	Nominal condition
Fusion-protein stock	50 µL	5.037 mg/mL stock
FITC stock	10 µL	10 mg/mL stock
Carbonate buffer	415 µL	0.1 M, pH 9.0
Glycine, after labelling	25 µL	1 M stock; 50 mM in 500 µL
Total volume	500 µL	As written in the source

Note. The stock-concentration calculation gives a nominal FITC:protein molar ratio near 17:1 when using 16,610.56 g/mol for the fusion; this is a theoretical reaction input, not labelling efficiency. The original August notes contain

inconsistent nanoparticle mass/concentration statements (including 0.01, 0.075 and 0.1 mg per condition), so the actual per-condition particle loading should be verified before reproducing the assay. Fluorescent fields were not consistently distinguishable from controls.





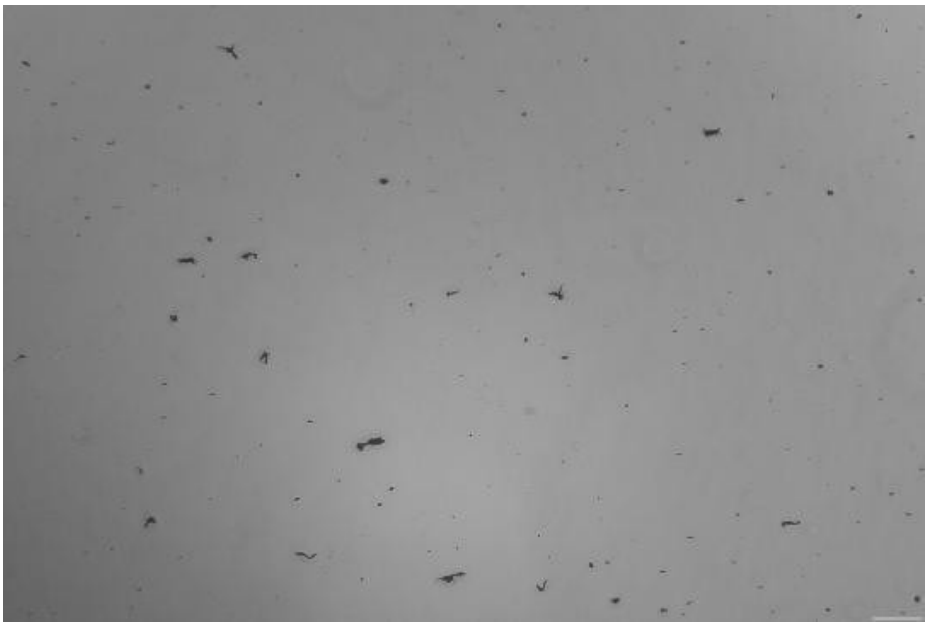


Figure. FITC-only/magnetic-particle control microscopy from the 4 August comparison.

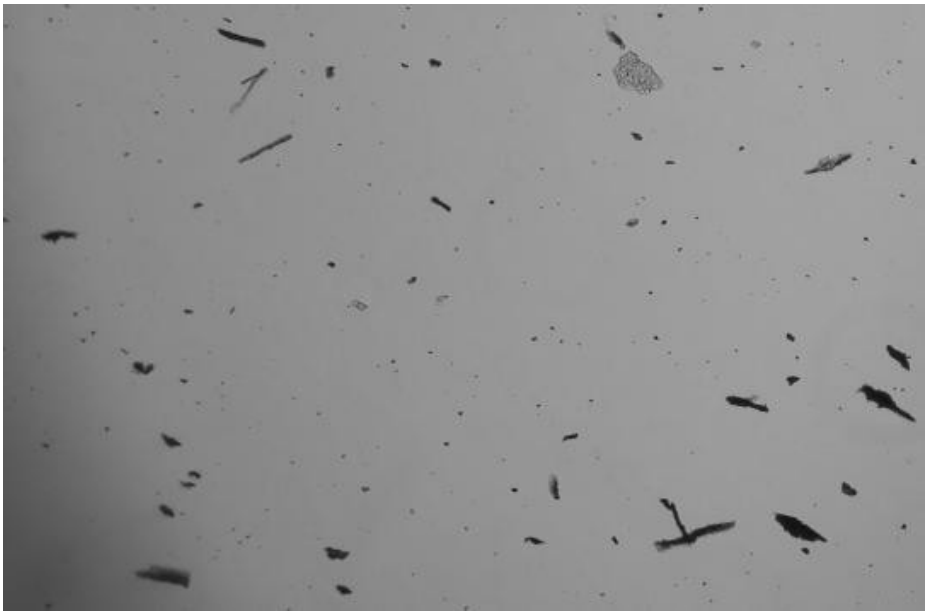


Figure. Representative microscopy of the fusion-protein preparation without glycine quench, with sonication.

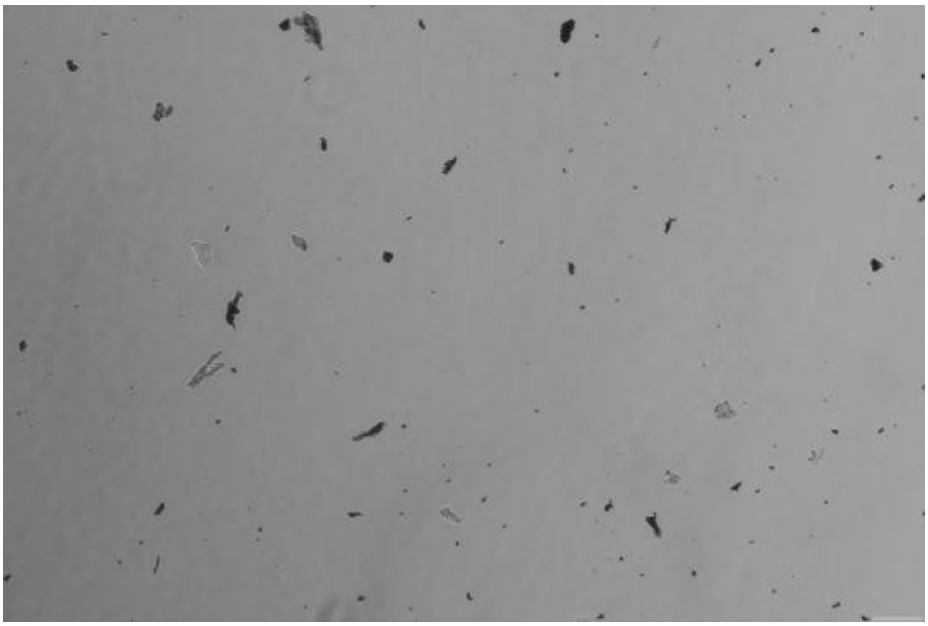


Figure. Microscopy of a quench/sonication comparison condition.

6 Repeat preparation and fluorescence follow-up

Date: 5–15 August 2026

Procedures

1. Samples were dialysed on 5–6 August, concentrated on 7 August and tested again by BCA and fluorescence on 15 August.

2. BCA measurements were performed on serially diluted samples (5-fold, 10-fold and 20-fold). The source regarded the 20-fold dilution as the preferred calculation range and estimated the concentration at 6.16 mg/mL after correction.

3. Another fluorescence assay was performed with a higher nominal TVN dose (recorded as “1000 µL/mL”, an ambiguous unit that is reproduced here rather than silently reinterpreted).

Note. The repeat fluorescence assay did not produce a convincing fluorescent signal. The original notes question whether the previously prepared FITC reagent had degraded. Sample dilution and labelling conditions were still under optimisation.

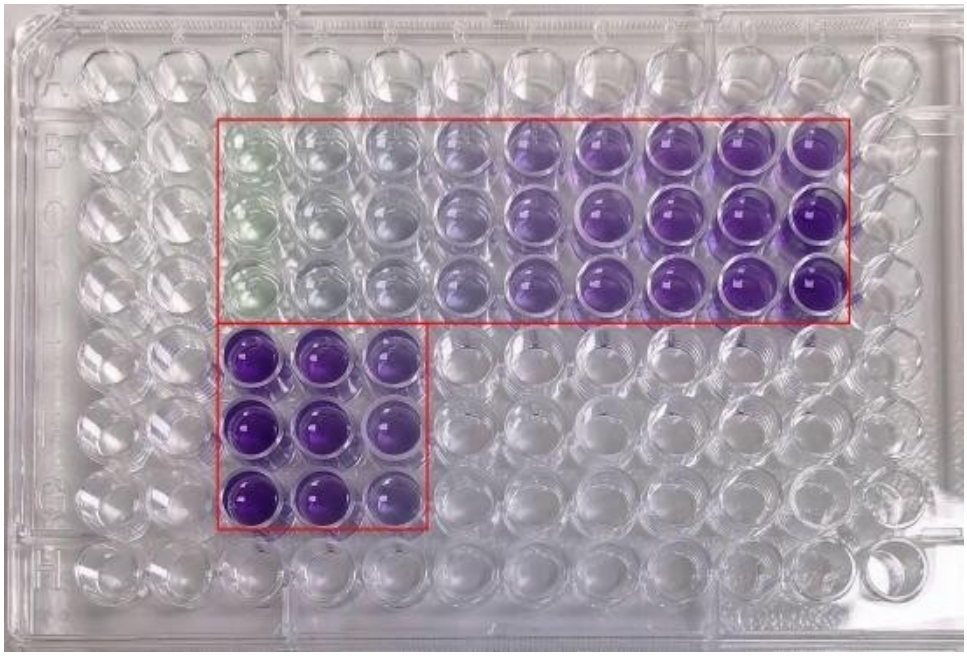


Figure. Repeat BCA quantification output associated with the August sample-preparation workflow.

7 Recovery from frozen stocks and renewed protein preparation

Date: 28 August–4 September 2026

Procedures

1. Direct streaking of the frozen glycerol stock on 28 August yielded no visible colonies; a second direct plating attempt on 29 August was also negative.

2. On 29 August, approximately 100 µL glycerol stock was transferred into 5 mL LB containing kanamycin to establish a liquid starter.

3. The overnight culture increased in OD and was expanded to an approximately 500 mL induction culture on 30 August. Samples were temporarily stored or frozen on 31 August.

4. Cells were harvested on 2 September and processed for resuspension, disruption and affinity purification on 3 September. The ultrafiltration operation on 3 September had a handling issue, and the result was excluded from later interpretation.

5. Further ultrafiltration and BCA analysis were recorded on 4 September, together with preparation of a separate CPC starter culture. The BCA data were assayed after 20-fold dilution.

Sample	BCA estimate (mg/mL)	Source interpretation
Before ultrafiltration	2.463	4 September result after ×20 dilution correction
After ultrafiltration	3.081	4 September result after ×20 dilution correction

Note. Liquid revival succeeded where direct plating failed, but no plating-efficiency count was recorded. The September protein concentration values represent a different batch from the July measurements and should not be pooled without matching sample IDs.

8 Revised fluorescence conditions and late-stage verification

Date: 16–28 September 2026

Procedures

1. On 16 September, a revised labelling plan retained the 0.1 M carbonate-buffer formulation (0.42 g Na_2CO_3 and approximately 0.10 g $NaHCO_3$ per 50 mL; adjusted to pH 9.0) and fresh FITC stock of 1.0 mg dissolved in 100 µL DMSO.

2. The Fe_3O_4 suspension was revised toward 0.075 mg/mL by combining 0.75 mL of a 1 mg/mL particle suspension with 9.25 mL water. The notes describe 0.075 mg particle input as the working comparison, though the mixture-volume/mass relationship was not fully reconciled.

3. The revised procedure called for 2 h dark FITC labelling, optional glycine termination, 2 h particle incubation in the dark with intermittent gentle mixing, magnetic separation and four to five gentle carbonate-buffer washes. The author questioned whether glycine termination should be retained for the particle-binding application.

4. A CPC comparison/pre-experiment was briefly recorded on 23 September; the source gives no experimental details sufficient for a separate functional conclusion.

5. On 27 September, FITC-labelled protein samples were examined by SDS-PAGE, and on 28 September fluorescence microscope imaging parameters were adjusted with follow-up images recorded.

Note. The final dates document additional protein/fluorescence verification activity, but the source does not contain a robust quantitative adsorption dataset, particle-bound/free-fluorophore balance, replicated binding kinetics or a definitive TVN-specific affinity value. The magnetic-particle interaction remains a qualitative, condition-sensitive observation.

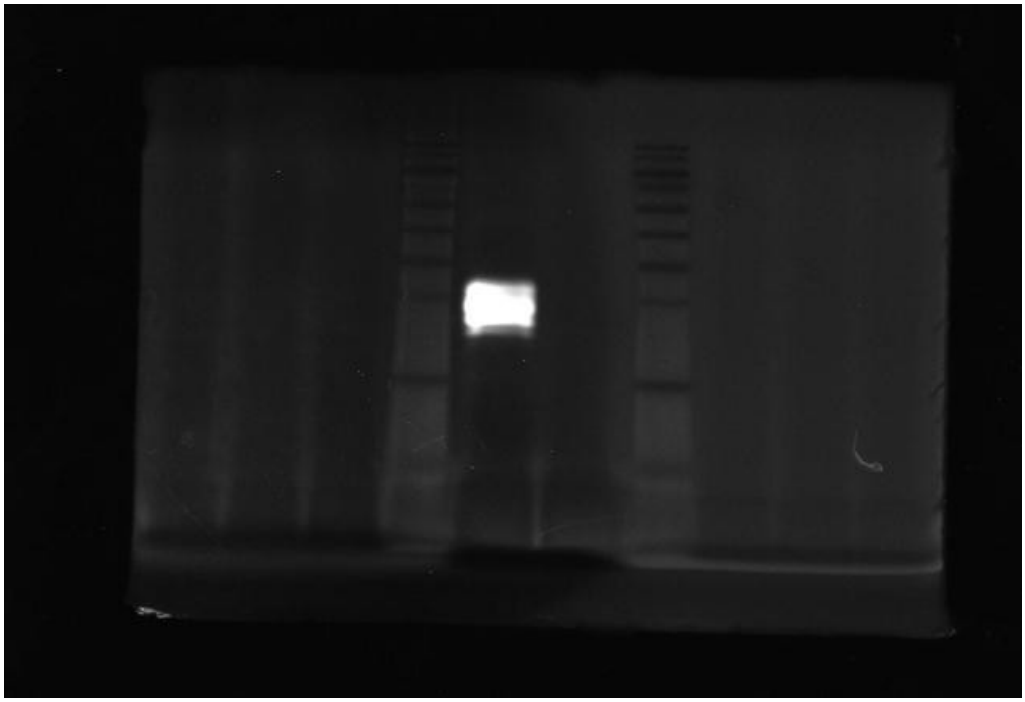


Figure. SDS-PAGE examination of fluorescently labelled samples recorded on 27 September.



Figure. Late-stage fluorescence imaging following microscope setting adjustments on 28 September.

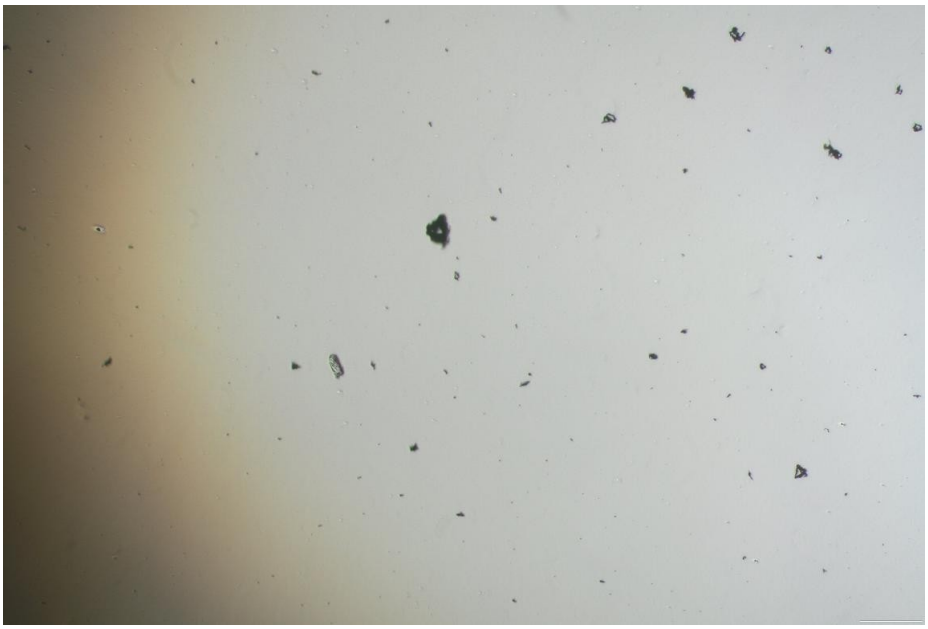


Figure. Additional fluorescence image from the 28 September follow-up.

Experimental interpretation and validation boundary

The work established a sequencing-supported TVN-containing expression construct, carried out recombinant fusion-protein production and concentration, and tested SnoopLigase2-associated DogTag coupling and fluorescence-based nanoparticle interaction. The first BCA depletion assays and subsequent FITC readouts were not sufficiently reproducible to establish a quantitative Fe_3O_4-binding performance claim. The high-molecular-weight SDS-PAGE bands observed in the ligation screens warrant further confirmation using clearly matched enzyme-negative controls. Future verification should use fresh FITC, matched fluorophore-only and nonbinding-peptide controls, fixed particle mass, normalised imaging exposures, replicated fluorescence measurements and a complementary unlabelled adsorption assay.

Referenced source materials: Mut Express II Fast Mutagenesis Kit V2 manual; SuperPlasmid Mini Kit manual; separate FITC–Fe_3O_4 peptide-binding protocol; archived Sanger traces; and Buldun, C. M., Jean, J. X., Bedford, M. R., & Howarth, M. (2018), Journal of the American Chemical Society 140, 3008–3018, DOI: 10.1021/jacs.7b13237. Some referenced files were named in the original notebook but were not supplied for independent inspection.